DILLON GREEN

FLIGHT SIMULATOR ENGINEER · SOFTWARE & AUTOMATION

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SUMMARY

Flight simulator engineer and self-taught software developer with 10 years in aviation, 8 of them hands-on with Boeing and Airbus avionics, full flight simulators and FAA-qualified training devices. Designed, built and deployed PAR, the ~72k-line parts, procurement and work-order platform used across the 40-person simulator department, and leads the intern-built MPIC test bench. Pairs hands-on simulator expertise with AI-assisted software development to modernize how operations teams work.

SOFTWARE PROJECTS

PAR (Parts & Requisitions Platform) — *Simulator Operations (sole developer)*

Python · Flask · SQLite · Bootstrap · GDAL · PyInstaller

- Designed and built the department's inventory, procurement, work-order and budget system: ~72k lines, 231 routes, 50+ tables, three user roles and a full audit ledger; rolled out to a 40-person department, in daily use.
- Migrated off legacy MS Access and GigaTrak with 15 importers, including a reviewable two-way reconcile engine (field-level diffs, learned vocabulary mapping, idempotent history import that keeps the original timestamps).
- Built a schematics pipeline that converts 6,000+ CAE MicroStation DGN drawings to searchable SVG via GDAL and links part numbers on drawings to live stock.
- Added fleet-level reorder logic across 26 simulator devices at two sites: spares aggregation, critical and low-stock tiers, ABC/FSN classification; plus a REST API, QR labels printed to Zebra printers, and mobile photo capture.
- Backed it with 340+ automated checks (72 regression suites, 142 headless-browser end-to-end checks, a 30-thread concurrency test), a ratcheting quality gate, automated offsite backups and a one-file Windows service deploy.

Linework — *Open-source CAD viewer and editor in the browser (linework.dillongreen.dev)*

JavaScript · Canvas · Web Workers · WebAssembly · pdf.js · Offline PWA · Playwright

- Opens DXF, DWG, MicroStation DGN and PDF drawings entirely on the user's device; measure, mark up, edit and save back to DXF or DGN.
- Traces a wire's whole net on a schematic in under 50 ms on a 53k-entity drawing; indexes a folder of sheets in the background, links off-sheet references and searches text across every sheet.
- Edits DGN files losslessly by patching the original bytes in place; reads DWG with LibreDWG compiled to WebAssembly in a bit-for-bit reproducible build and loaded only on demand, keeping app startup under a 150 KB budget; 330+ automated unit and end-to-end tests.

Cookfold — *Recipe and meal-planning app for iPhone (live on the App Store; formerly Crumb)*

JavaScript · Capacitor · Swift · RevenueCat · Cloudflare · Codemagic CI

- Wrote a native Swift plugin for on-device Vision OCR, iCloud sync and a share extension; local-first sync design; subscriptions; shipped through App Store review with Codemagic CI builds to TestFlight; 460+ automated checks.

pocket429 — *ARINC 429 decoder for avionics technicians (iOS app in App Store review; pocket429.dillongreen.dev)*

JavaScript · Capacitor · Cloudflare Workers · Stripe

- Decodes and builds ARINC 429 data words (BNR, BCD, ISO-5, parity, bit-order detection) against a label database compiled from cited OEM manuals.

Dragon Realm Online — *Real-time multiplayer browser game (play)*

JavaScript · Canvas · Cloudflare Durable Objects · WebSockets · D1

- Live position sync, chat, trading and duels over hibernating WebSockets, with a player market stored in D1 (~11k lines).

Little Airfield — *Cozy airport-builder game for iPhone (playable prototype)*

JavaScript · Canvas (isometric, art drawn in code) · Playwright

- Clear planes to land into the wind, fix pilot-reported squawks and grow a grass strip into a towered airport; pure rules engine with automated tests and iPhone-landscape end-to-end checks.

Also built: Smart Mirror (Raspberry Pi magic mirror), Mixtape Drift (rhythm street racer with a WebAudio synth band), surprisepass (live, Stripe payments), halfsies (receipt-OCR bill splitter), and butter and beepbeach (canvas games). Portfolio: dillongreen.dev.

EXPERIENCE

Alaska Airlines — Seattle, WA

Flight Simulator Engineer — **June 2025 – Present**

- Engineer for Boeing 737, 787 and Airbus A330 full flight simulators on CAE XR, R4, R5 and L3Harris platforms, holding each training device (FSTD) to FAA qualification standards under 14 CFR Part 60, including running and supporting QTG testing.
- Lead the summer intern team building a standalone MPIC test bench: 16 MPICs in a dedicated I/O chassis, driven by a repurposed Windows Server 2019 sim node on its own 48-port network. The bench tests and calibrates I/O and checks aircraft panels from CAE's OneUI without taking a training device offline; loading the aircraft simulation onto it is the next phase.
- Have taken part in numerous simulator acceptances, FAA evaluations and initial qualifications, including in-plant acceptance at CAE in Montreal.
- Write code-level fixes across simulator platforms for faults that hardware or configuration changes can't resolve.
- Primary troubleshooter for IOS and interface faults (cockpit I/O to the simulation). Lead interface modifications end to end: hardware, cabling, drawings and interface software packages, including the 2KU ground control panel on five simulators.
- Built and deployed PAR, the department's parts, procurement and work-order platform (see Software Projects).
- Fly the 737, 787 and A330 simulators for the daily preflight check, and work with regulatory and pilot-training teams to sustain FAA-mandated qualification.
- Work rotating shifts in a 24/7 training operation, troubleshooting across interface, IOS, motion and visual systems.

Flight Simulator Technician II / III — **Feb 2022 – June 2025**

- Maintained and troubleshot CAE and L3Harris full flight simulators, clearing time-critical faults before scheduled training sessions.
- Integrated new hardware through MPIC interface management; primary troubleshooter for ISS/IOS issues.
- Resolved recurring discrepancies from schematics, wiring diagrams and pilot handbooks; kept FAA qualification records.

Panasonic Avionics — *Maintenance Services Technician* — 2018 – 2022

- Performed systems verification and in-flight entertainment software validation inside Boeing 737 MAX and 787 production facilities ahead of customer delivery.
- Worked on aircraft at Seattle-Tacoma International Airport (SEA), diagnosing complex faults across Boeing 737, 757, 767, 787 and Airbus A320 and A330 fleets and running software installs and end-to-end system checkouts in live airline operations.
- Supported engineers on electrical troubleshooting using wiring diagrams and system schematics.

SkyWest Airlines — *Parts Clerk I (2018); Cross-Utilized Agent (2016–2018)* — 2016 – 2018

- Managed parts inventory and sourcing for CRJ and ERJ line maintenance; earlier, ramp, gate and customer-service operations.

SKILLS

- **AI & tooling:** AI-assisted development and code review (Claude Code, Anthropic API)
- **Languages:** Python, JavaScript, Swift, SQL, PowerShell; working knowledge of C++, C#, Fortran
- **Platforms:** Flask, SQLite/PostgreSQL, Cloudflare Workers, Capacitor (iOS & Android), RevenueCat, GDAL
- **Engineering practice:** automated regression and headless-browser testing, CI (Codemagic), data migration and reconciliation, backup/restore, Windows service deployment
- **Simulation & aviation:** CAE XR/R4/R5 and L3Harris full flight simulators; MPIC interface and ISS/IOS; working knowledge of motion and visual systems; Boeing 737/787 and Airbus A320/A330 systems; ARINC 429; electrical schematics and vendor drawings; FAA simulator qualification (14 CFR Part 60, QTGs)
- **Certification:** CAE XR Maintenance Certificate